Question ID: 0045c234

Question

Given that stars and planets initially form from the same gas and dust in space, some astronomers have posited that host stars (such as the Sun) and their planets (such as those in our solar system) are composed of the same materials, with the planets containing equal or smaller quantities of the materials that make up the host star. This idea is also supported by evidence that rocky planets in our solar system are composed of some of the same materials as the Sun.

Which finding, if true, would most directly weaken the astronomers' claim?

Answer

A. Most stars are made of hydrogen and helium, but when cooled they are revealed to contain small amounts of iron and silicate.

B. A nearby host star is observed to contain the same proportion of hydrogen and helium as that of the Sun.

C. Evidence emerges that the amount of iron in some rocky planets is considerably higher than the amount in their host star.

D. The method for determining the composition of rocky planets is discovered to be less effective when used to analyze other kinds of planets.